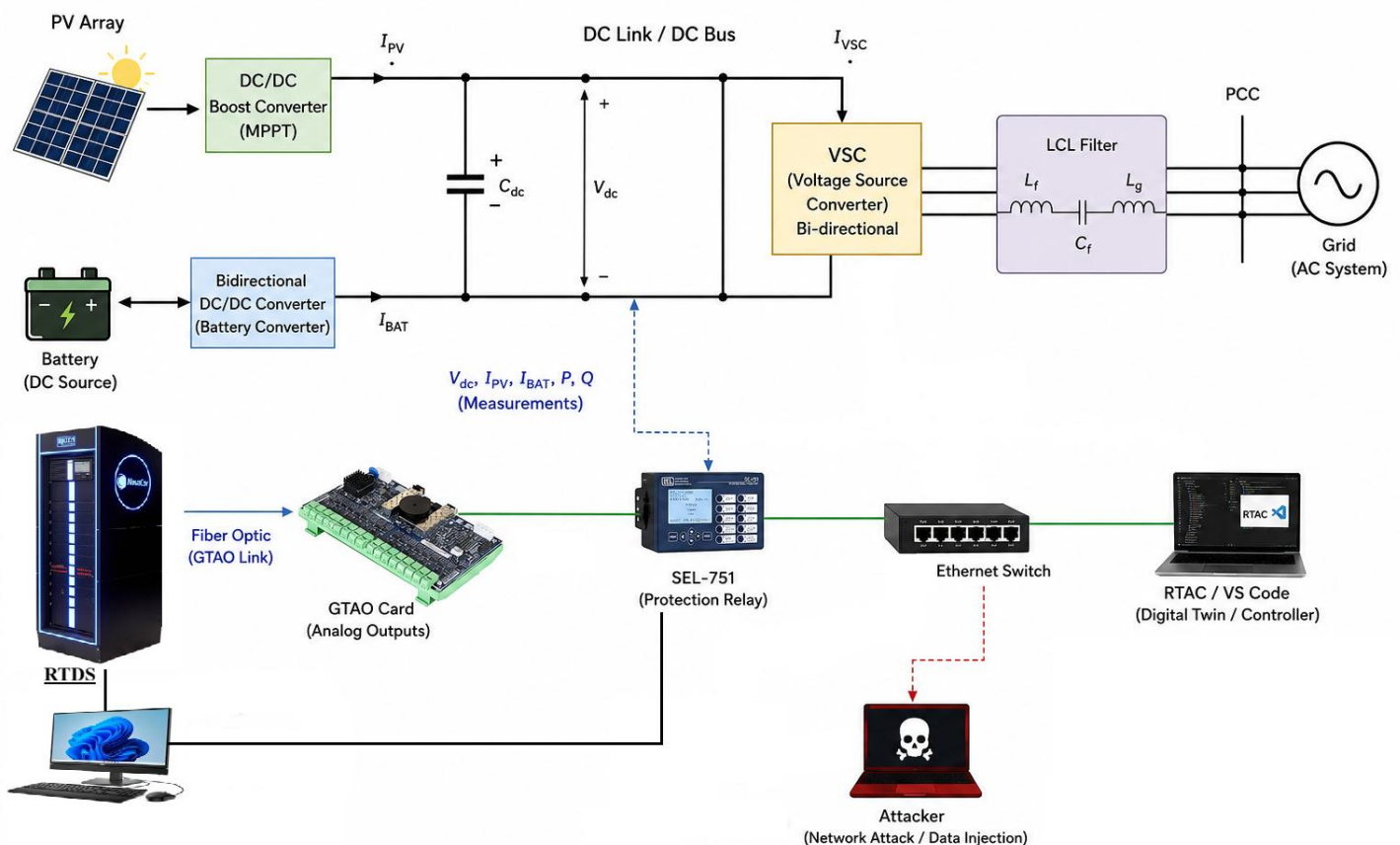


Cyber Attack Detection in a Grid-Connected PV–Battery System Using RTDS, SEL-751 Relay, RTAC, and Modbus Communication

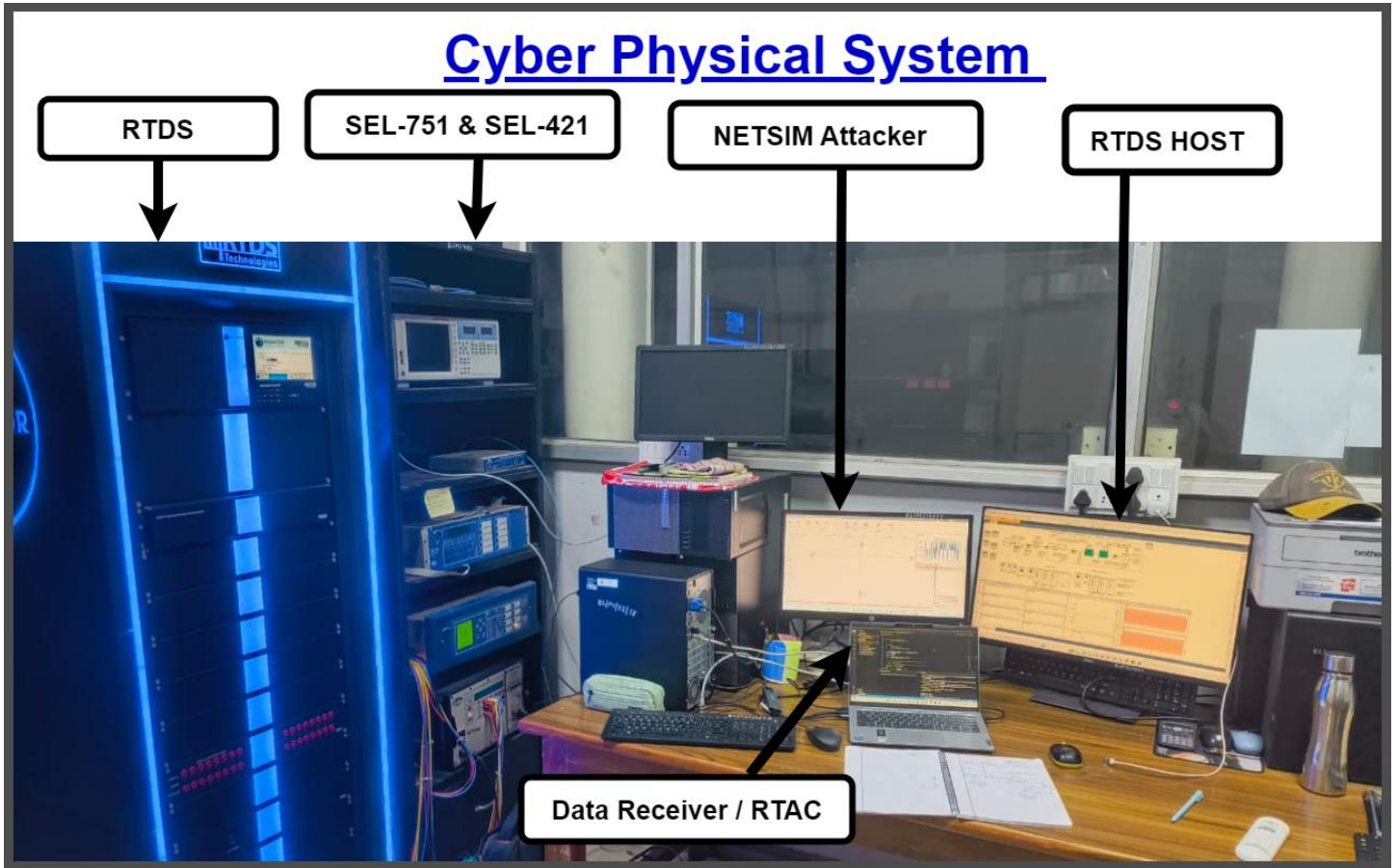
System details: -

The proposed project presents a **cyber-physical testbed** for detecting cyber attacks in a **grid-connected photovoltaic (PV) and battery energy storage system** using a Real-Time Digital Simulator (RTDS), SEL-751 protection relay, and Real-Time Automation Controller (RTAC). The complete electrical power system, including the PV array, battery energy storage system (BESS), DC-link, bidirectional Voltage Source Converter (VSC), LCL filter, and utility grid, is modeled and executed in the RTDS environment. The RTDS continuously generates real-time measurements such as DC-link voltage, PV current, battery current, active power, and reactive power.



These analog measurements are transmitted from the RTDS to the **SEL-751 relay** through the **GTA0 (GT Analog Output) card** using fiber-optic communication. The SEL-751 relay acquires and processes the measurements and acts as the field protection device within the cyber-physical system. The measured electrical parameters are then communicated from the SEL-751 relay to the **RTAC/Visual Studio Code (VS Code)** platform using the **Modbus TCP communication protocol**. The RTAC functions as the data acquisition and supervisory controller, while the cyber attack detection algorithm is implemented in the VS Code environment.

The proposed cyber attack detection technique continuously monitors incoming measurement data and compares it with predefined operating limits or expected system behavior. Under normal operating conditions, the measured signals remain within acceptable thresholds. However, during cyber attacks such as **false data injection (FDI)**, **replay attacks**, or **data manipulation**, abnormal deviations in voltage, current, or power measurements are detected. Once an attack is identified, the detection algorithm immediately generates an alarm signal and transmits a **trip command** to the SEL-751 relay through the **GOOSE (Generic Object-Oriented Substation Event)** communication protocol. The SEL-751 relay then isolates the affected system by issuing a protection trip command, preventing the propagation of faulty or malicious operating conditions.



The proposed Controller Hardware-in-the-Loop (CHIL)-based cyber-physical platform enables real-time evaluation of cyber attack scenarios and protection strategies without requiring a physical power system. This architecture provides a secure, flexible, and cost-effective environment for validating cyber-security algorithms, communication protocols, and protection schemes for modern smart grids and renewable energy systems.

Mathematical Model for Attack Detection:-

The detection algorithm compares the measured signal with its expected value:

$$e(t) = x_{measured}(t) - x_{expected}(t)$$

where:

- $x_{measured}$ = received measurement from SEL-751
- $x_{expected}$ = estimated or reference value
- $e(t)$ = detection error

An attack is declared when

$$|e(t)| > \delta$$

where δ is the predefined threshold.

For example,

$$|V_{dc}^{measured} - V_{dc}^{reference}| > \delta_V$$

or

$$|p^{measured} - p^{estimated}| > \delta_P$$

If either condition is satisfied for a specified time interval, the RTAC/VS Code identifies the event as a cyber-attack and transmits a **GOOSE trip command** to the SEL-751 relay to disconnect the affected section.

Conclusion: -

The proposed cyber-physical platform combines RTDS, SEL-751, RTAC, and Modbus communication to detect cyber-attacks in a grid-connected PV–battery system. The attack detection algorithm implemented in VS Code identifies abnormal measurement deviations and issues a high-speed GOOSE trip command to the SEL-751 relay, providing reliable and real-time protection against false data injection and communication-based cyber-attacks.